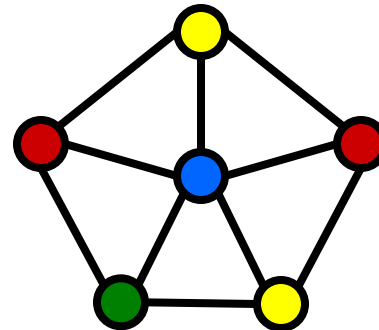
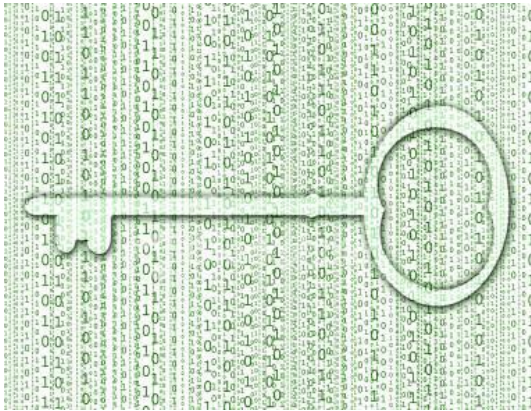
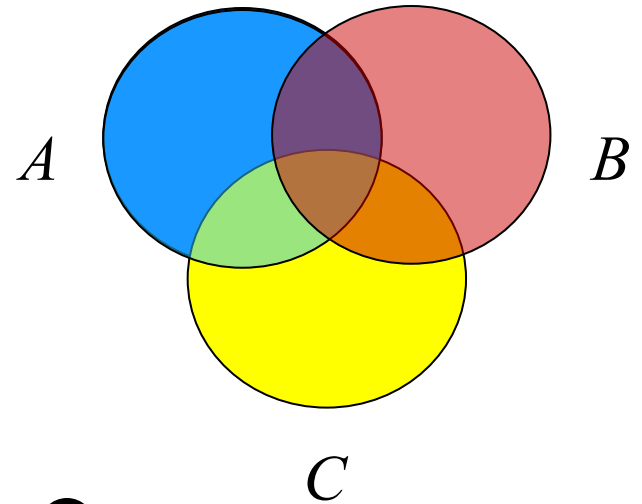


Introduction to Discrete Mathematics

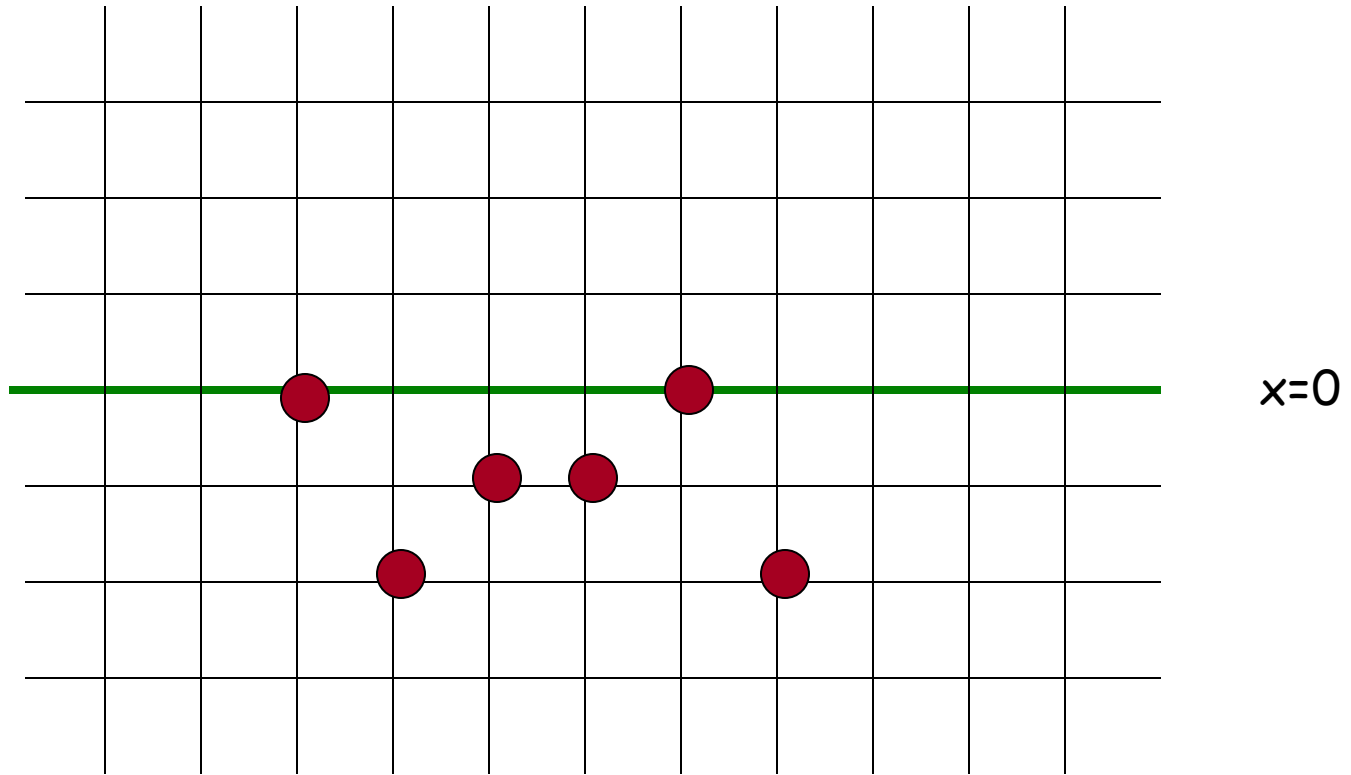
$$\frac{x_1 + x_2 + \dots + x_n}{n} \geq \sqrt[n]{x_1 \cdot x_2 \cdots x_n}$$



$$a = qb + r \implies \gcd(a, b) = \gcd(b, r)$$

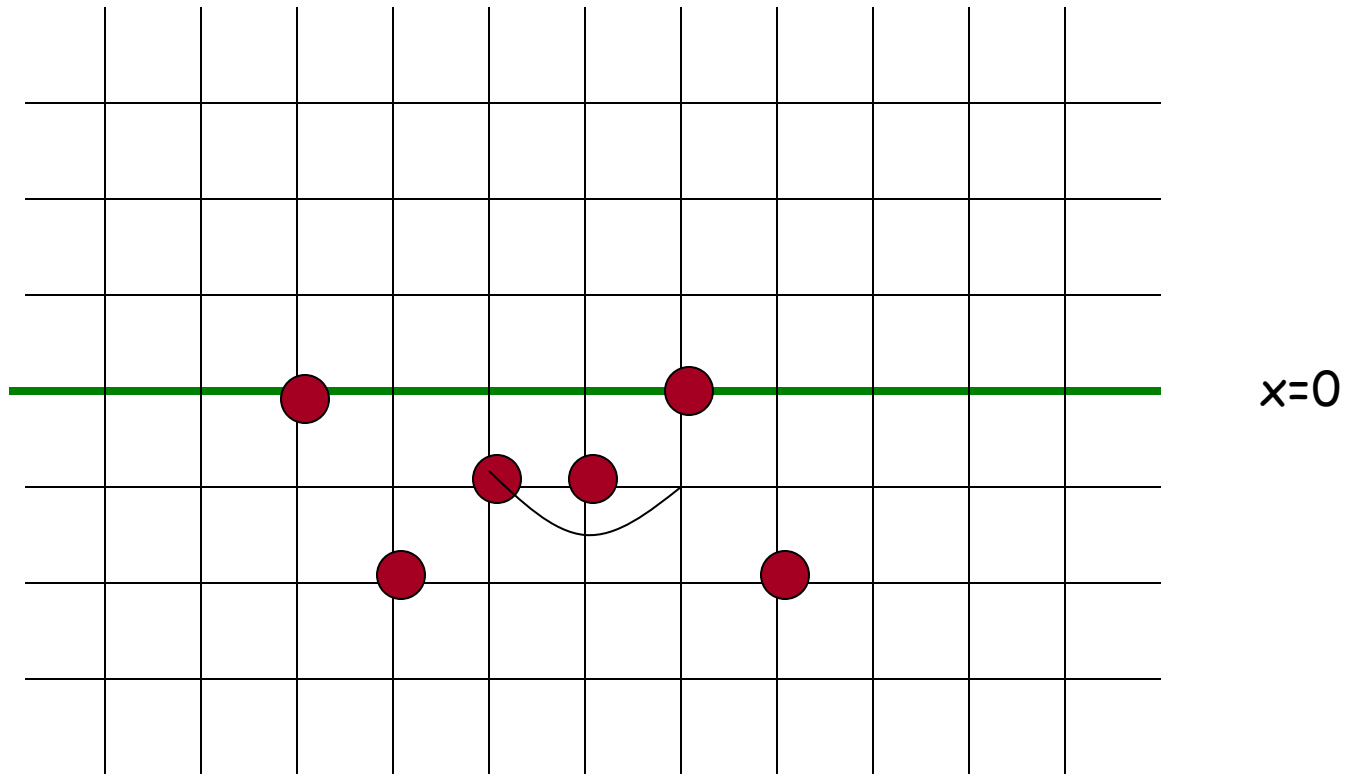


Checker



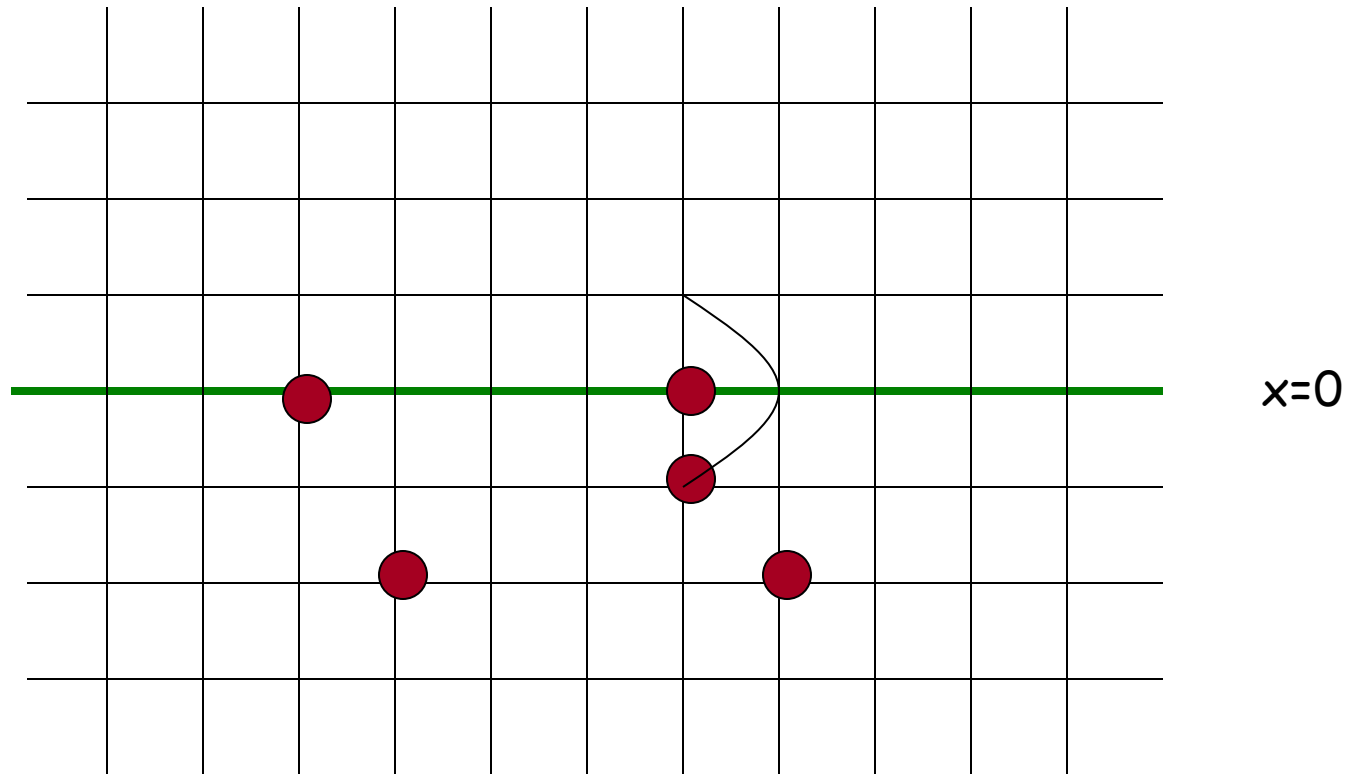
Start with any configuration with all circles on or below the x -axis.

Checker



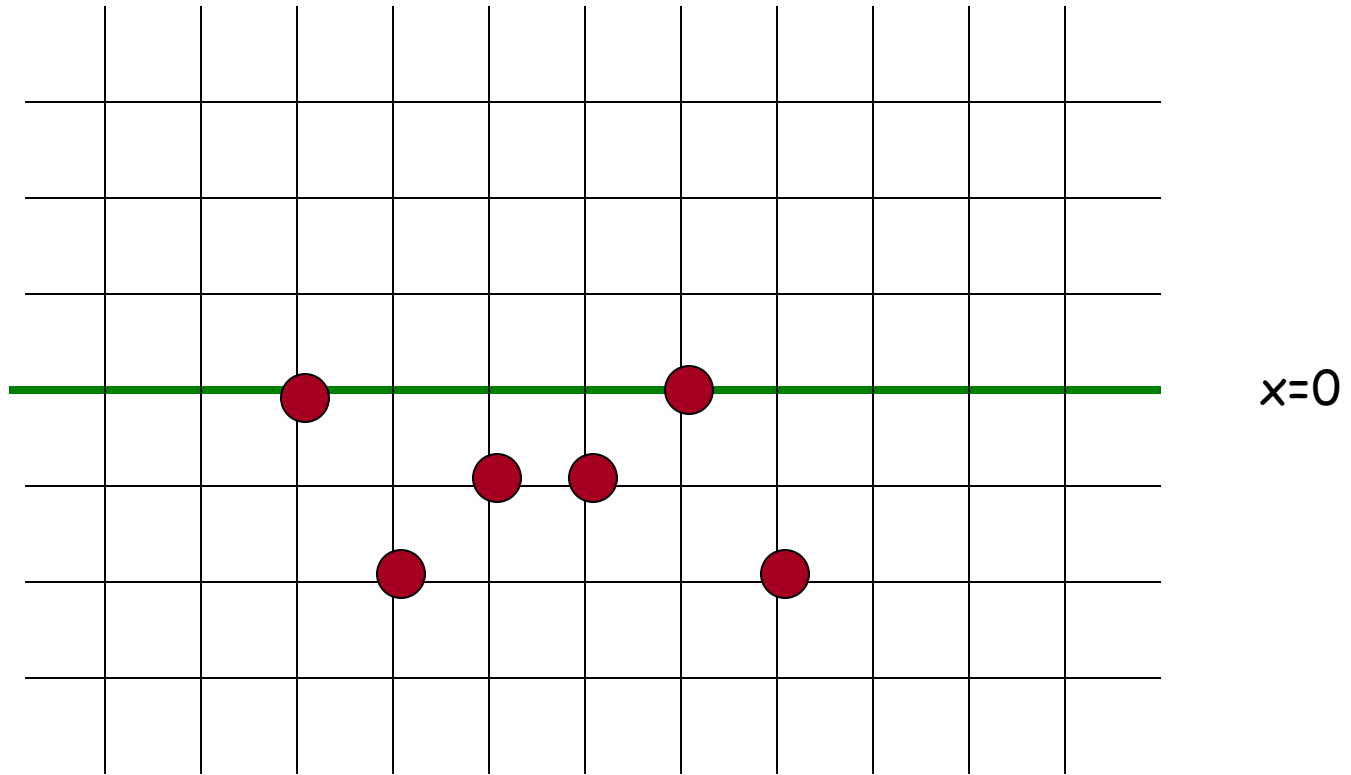
Move: jump through your adjacent neighbour,
but then your neighbour will disappear.

Checker



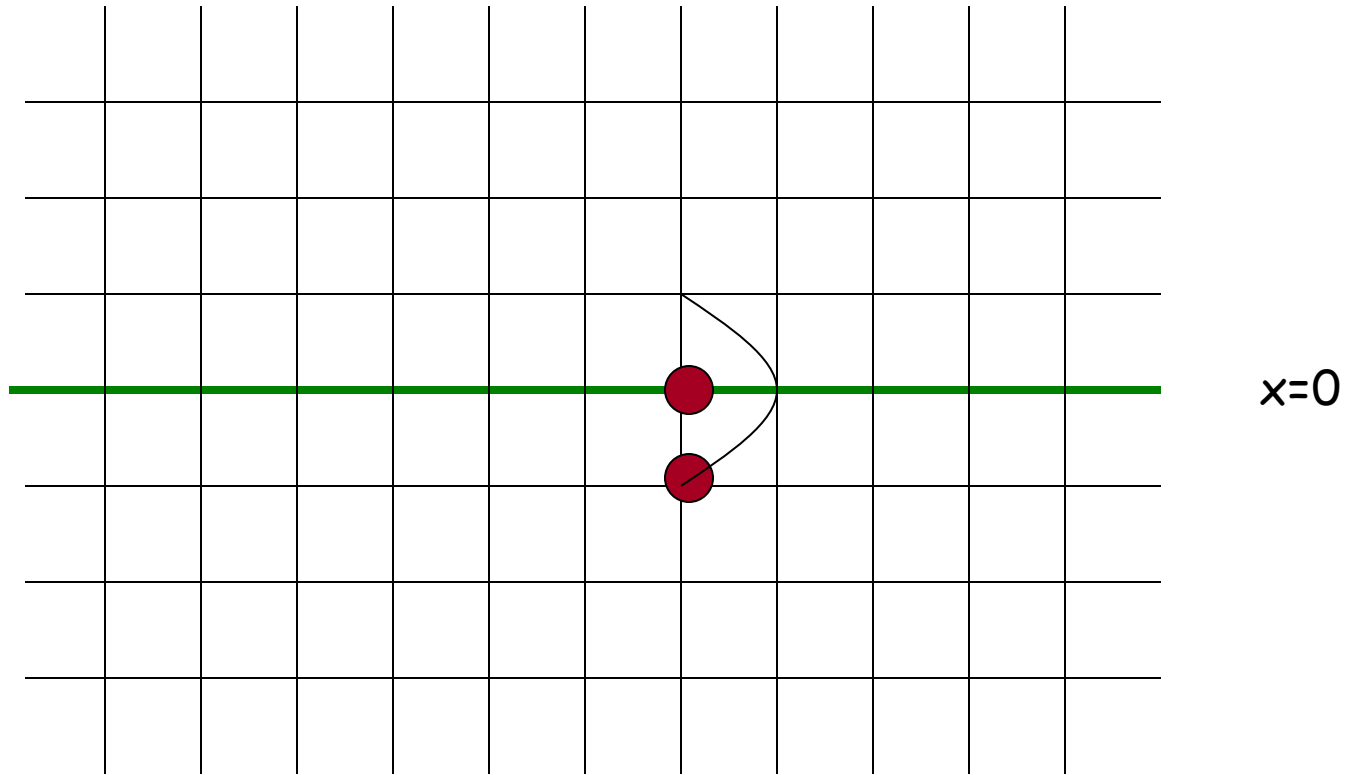
Move: jump through your adjacent neighbour, but then your neighbour will disappear.

Checker



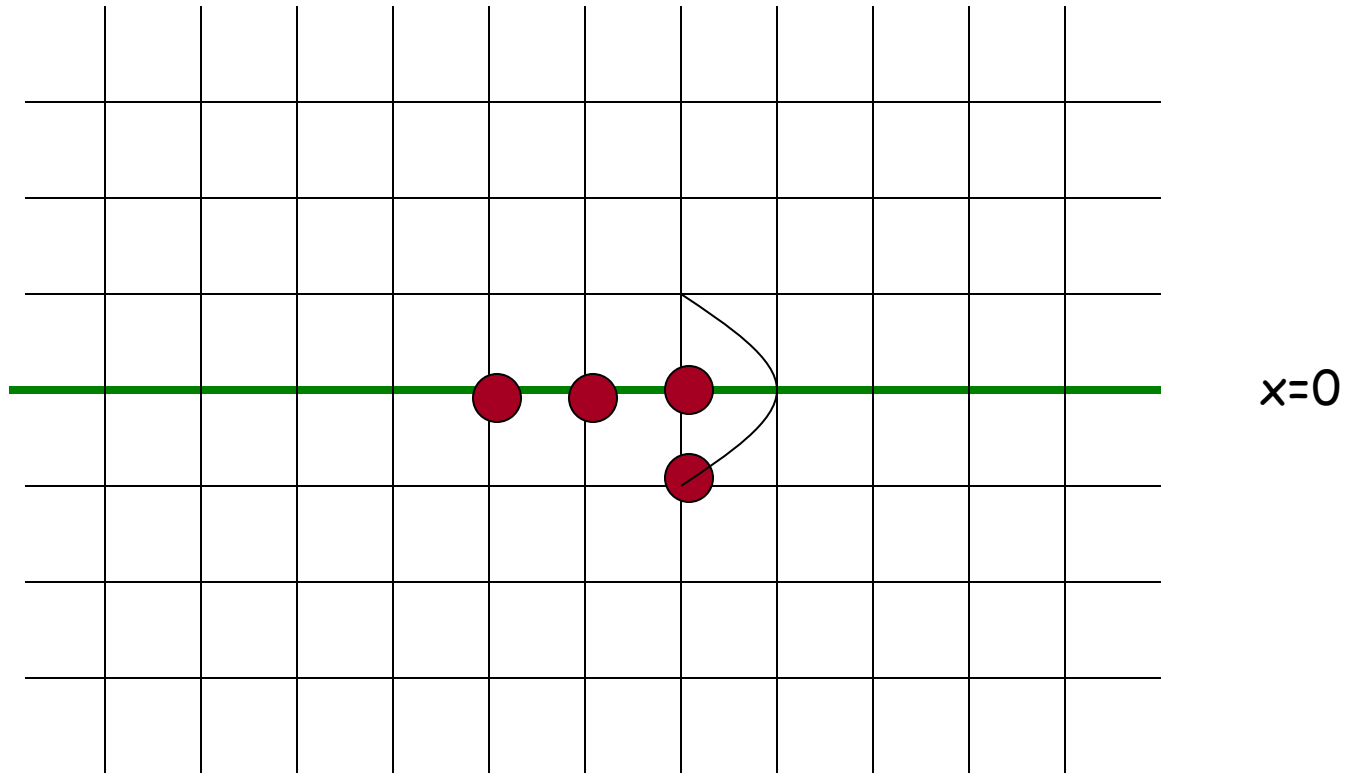
Goal: Find an initial configuration with **least** number of circles to jump up to level k.

$K=1$

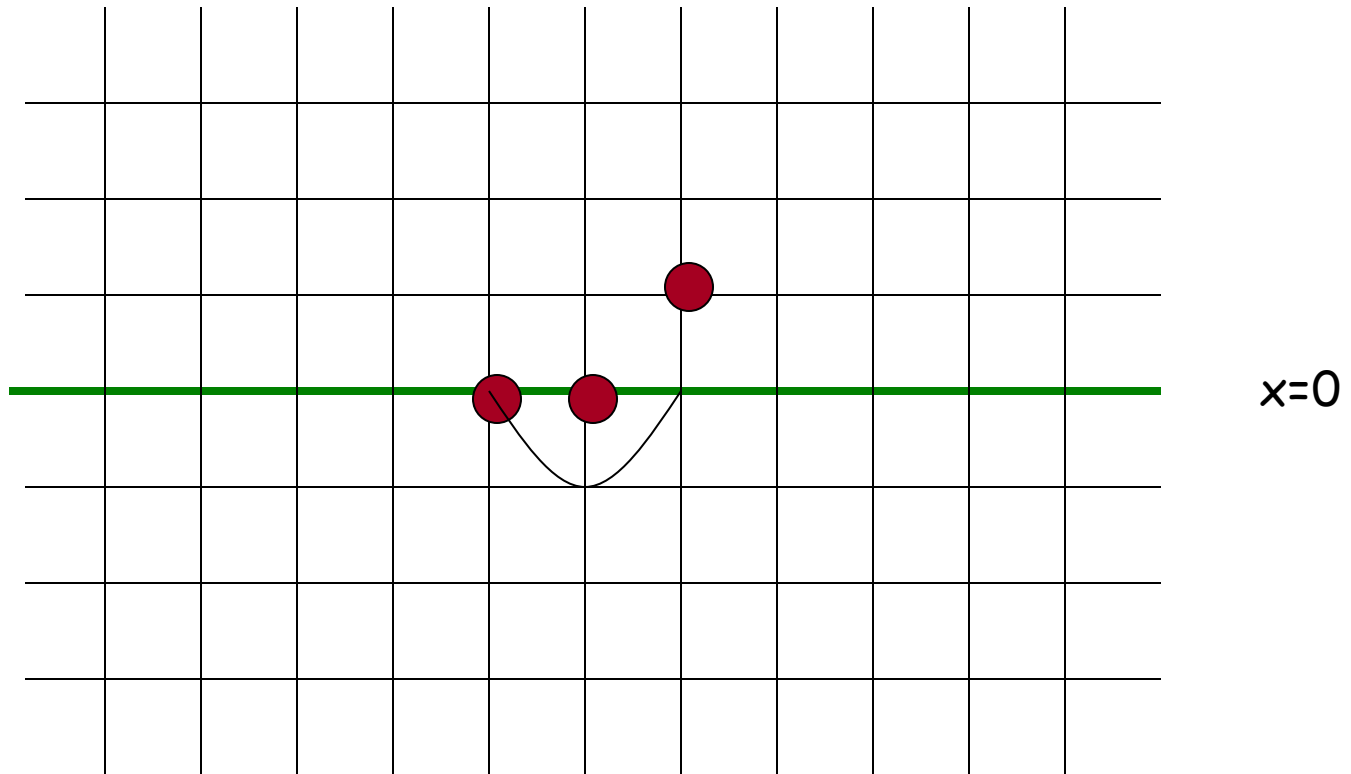


2 circles to reach level 1.

$K=2$



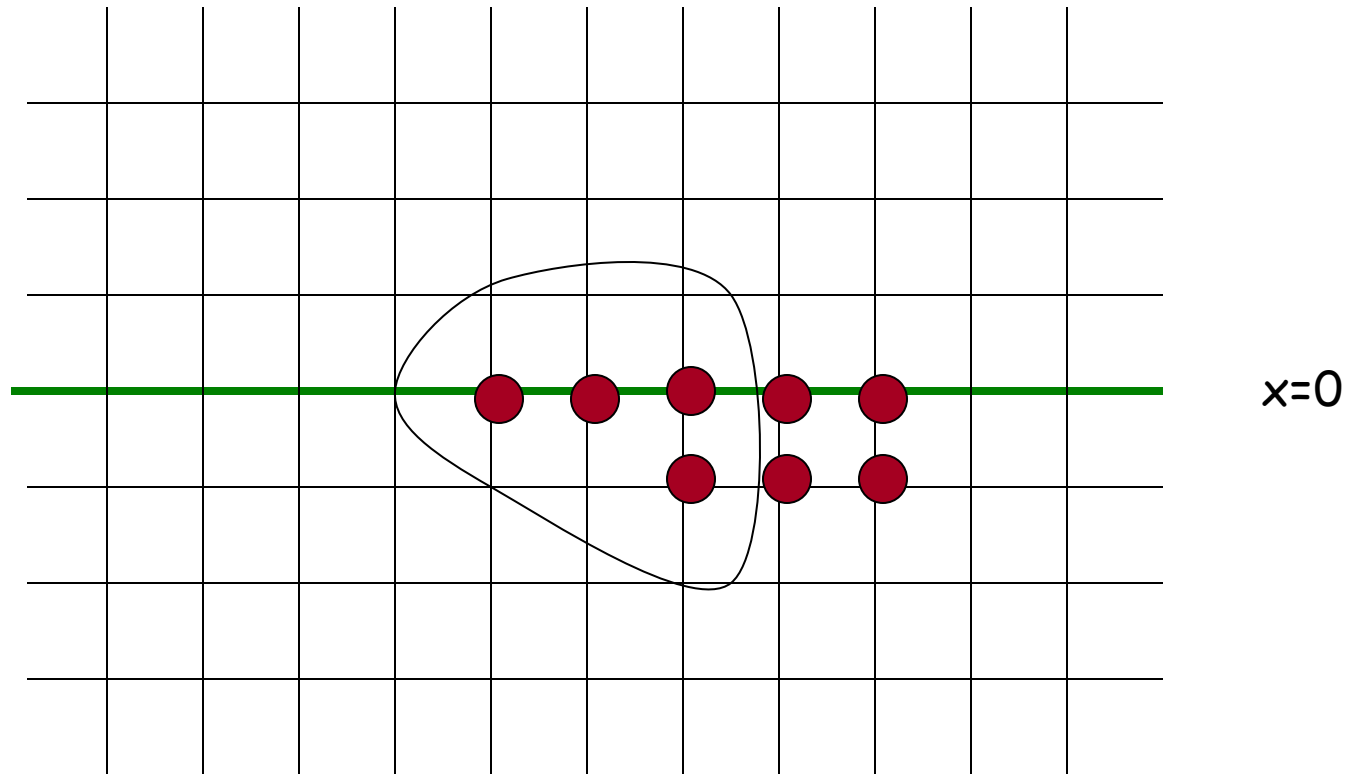
$K=2$



Now we have reduced to the $k=1$ configuration, but one level higher.

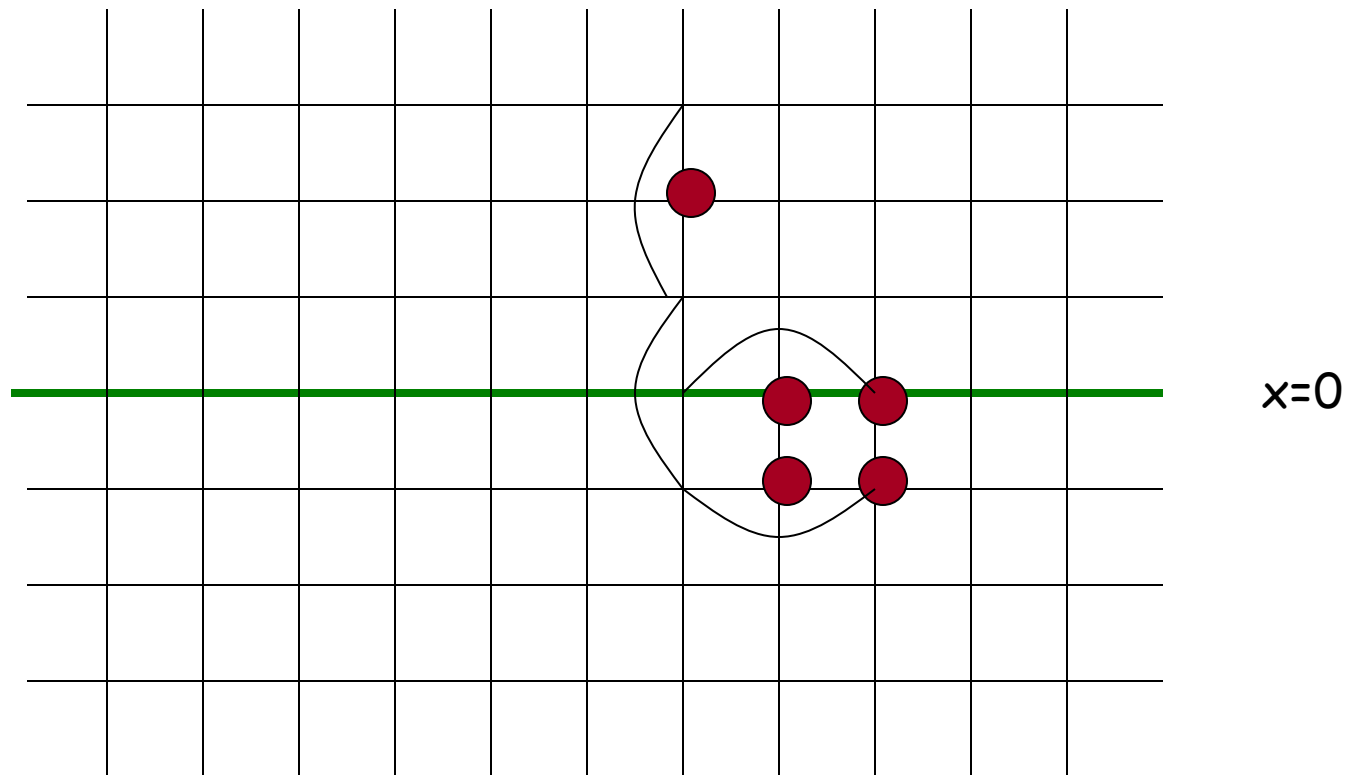
4 circles to reach level 2.

$K=3$



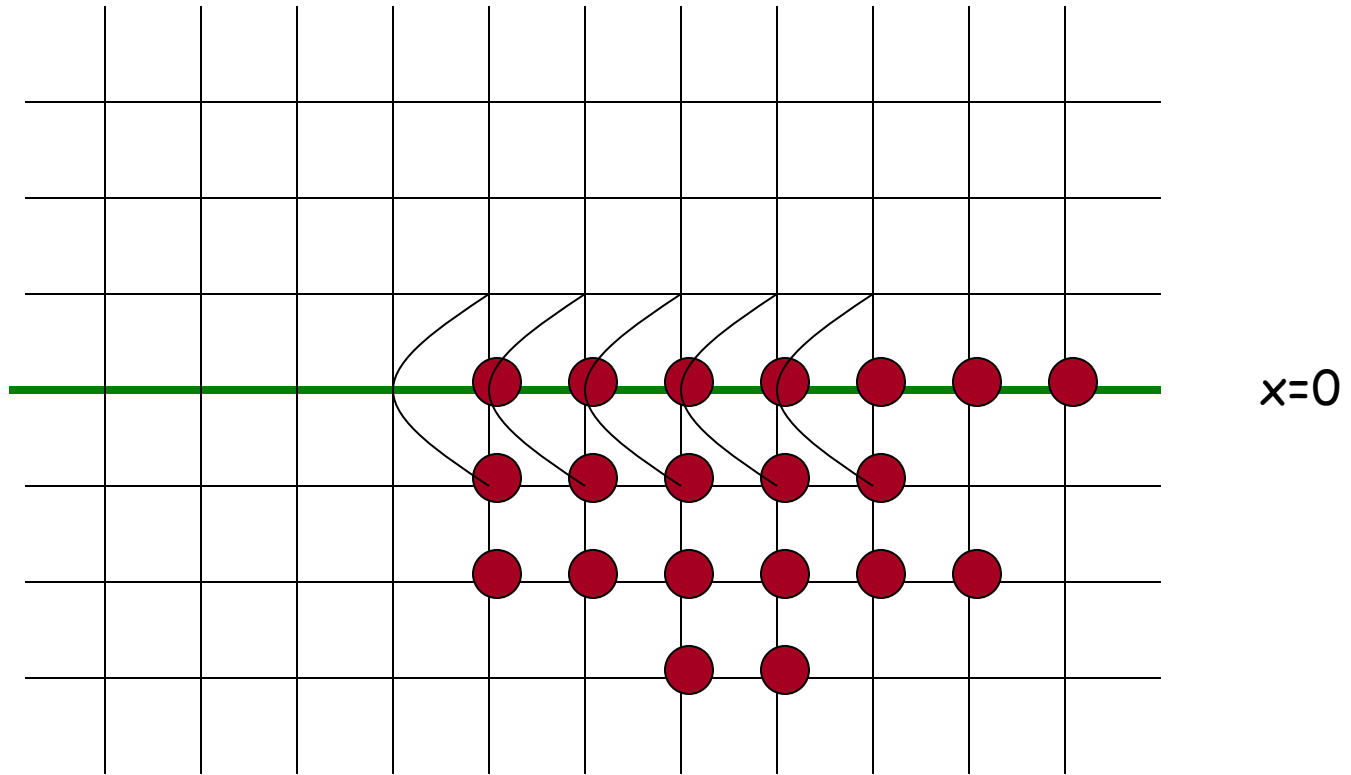
This is the configuration for $k=2$, so jump two level higher.

K=3

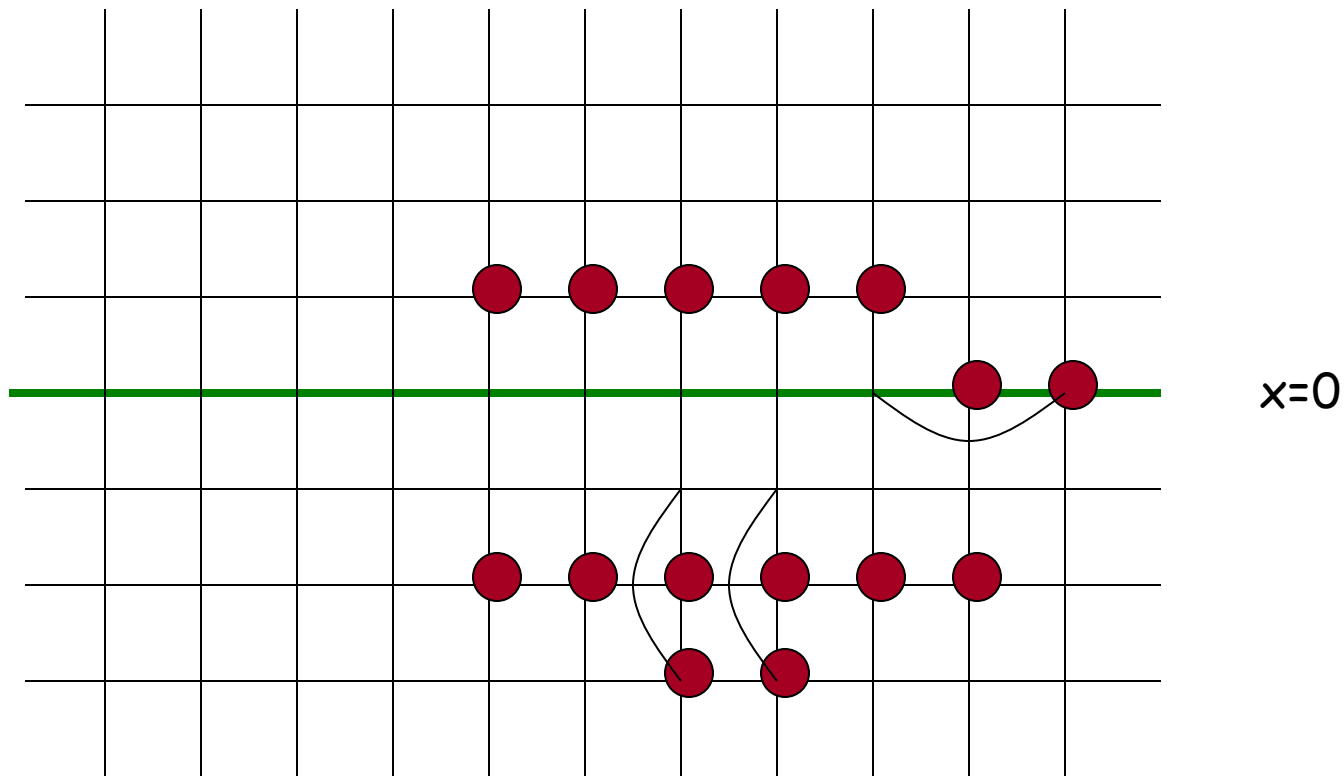


8 circles to reach level 3.

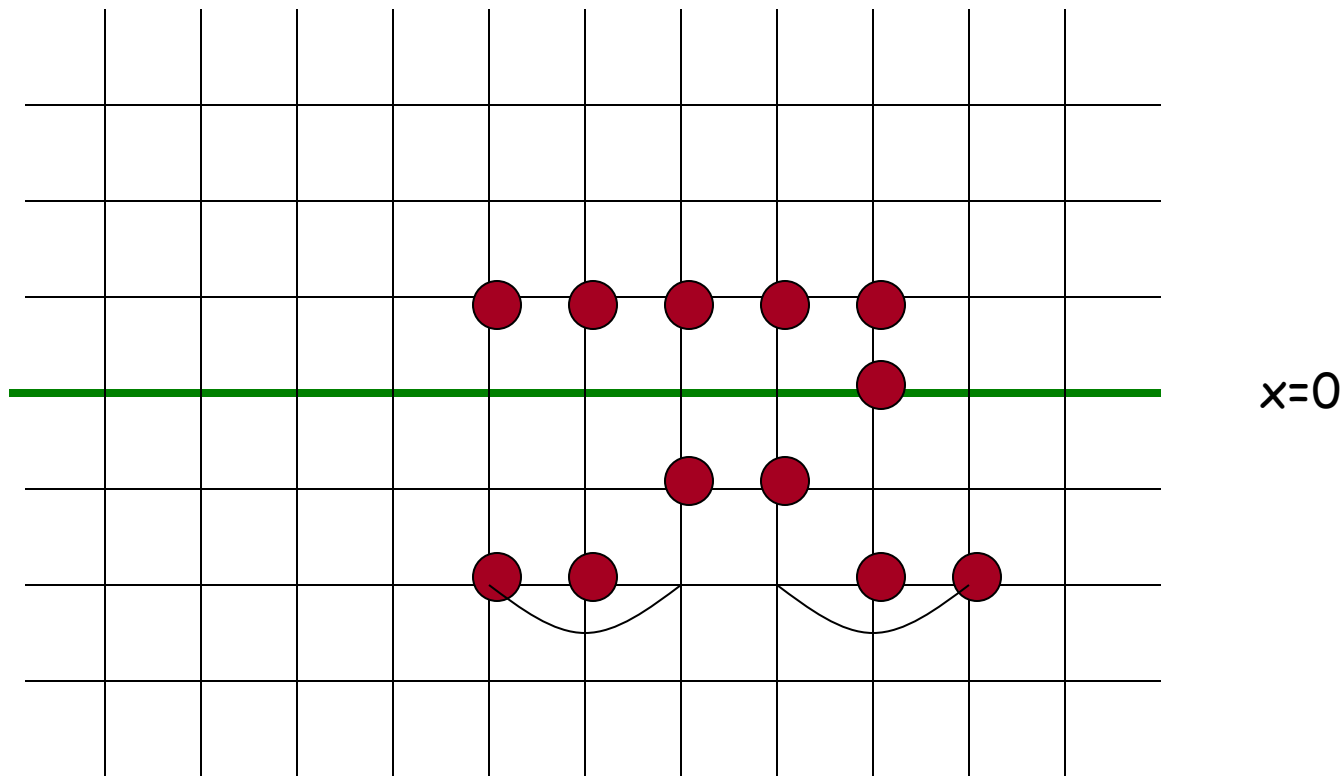
$K=4$



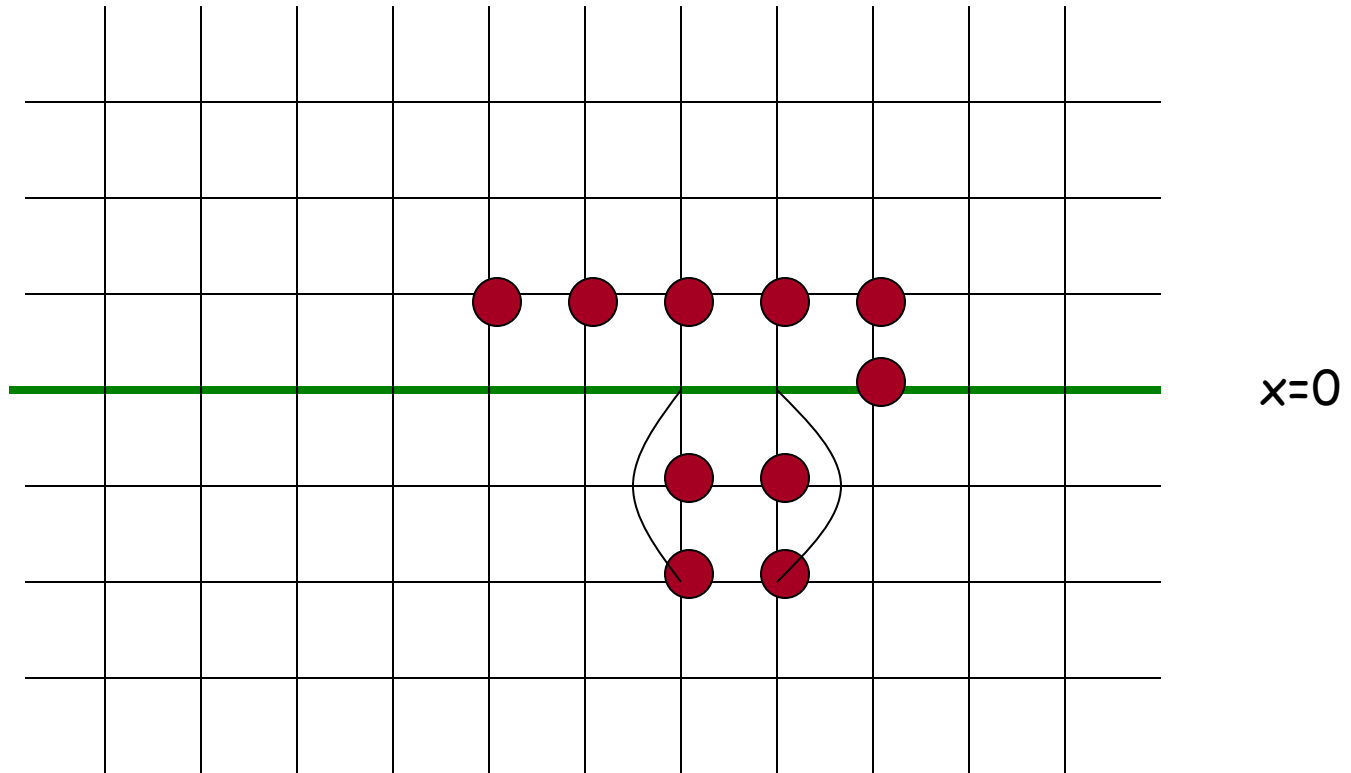
$K=4$



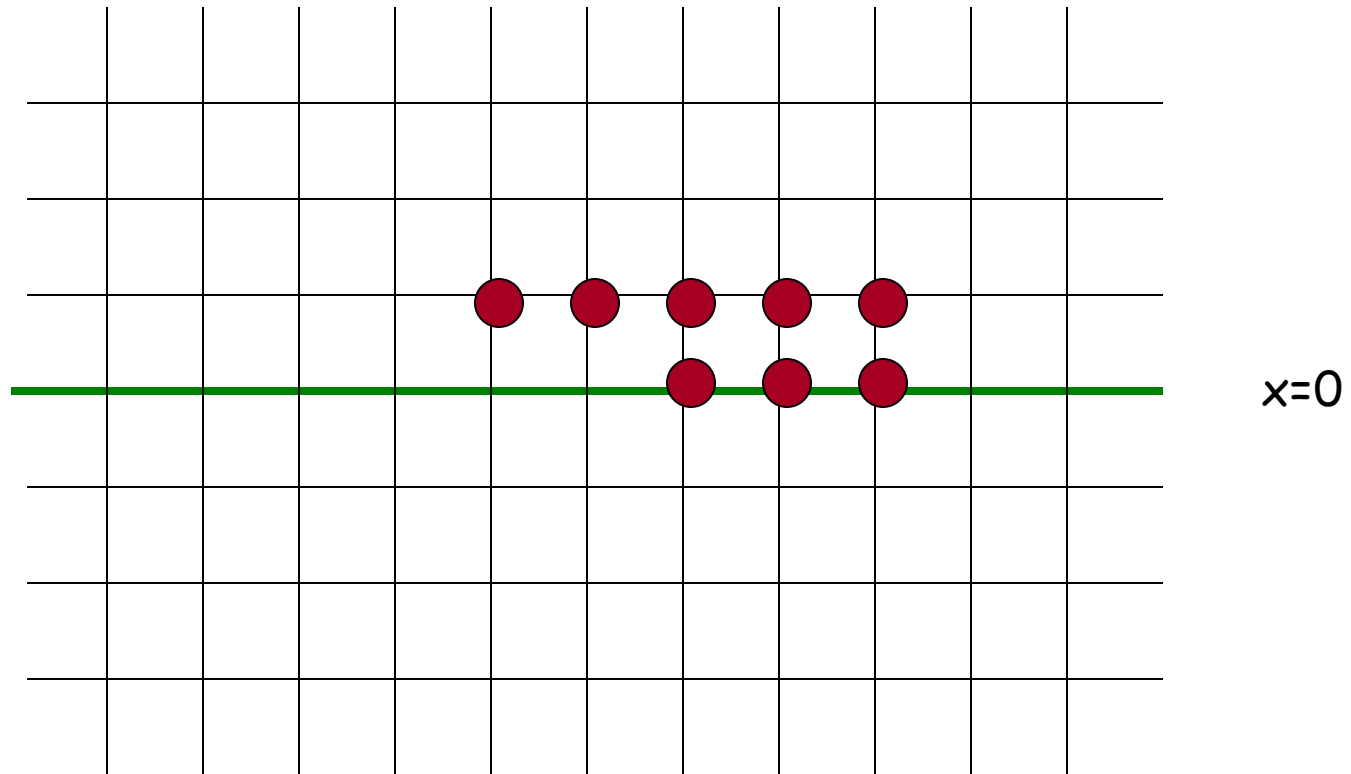
$K=4$



$K=4$



$K=4$



Now we have reduced to the $k=3$ configuration, but one level higher

20 circles (not 16) to reach level 4!

K=5?

- a. 39 or below
- b. 40-50 circles
- c. 51-70 circles
- d. 71- 100 circles
- e. 101 - 1000 circles
- f. 1001 or above

It turns out that it is impossible to move to level 5,
and there is a very interesting mathematical proof of it.



Why Discrete Mathematics?

Understand the design of efficient computer systems.

- How did Google manage to build a fast search engine?
- What is the foundation of internet security?

algorithms, data structures, database,
parallel computing, distributed systems,
cryptography, computer networks...

Logic, number theory, counting, graph theory...

Topic: Logic and Proofs

How do computers (and humans) think?

Logic: propositional logic, first order logic

Proof: induction, contradiction

$$\frac{x_1 + x_2 + \dots + x_n}{n} \geq \sqrt[n]{x_1 \cdot x_2 \cdots x_n}$$

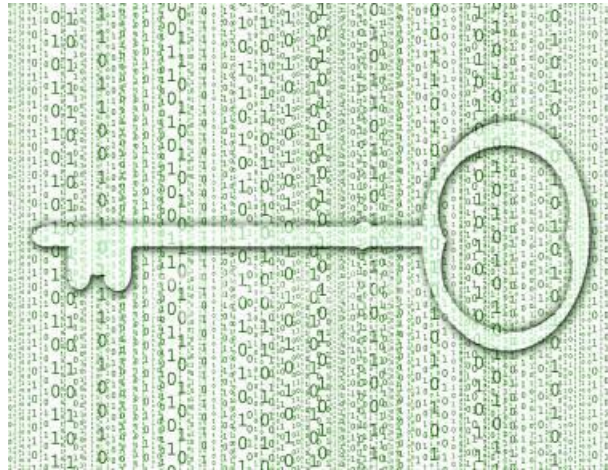
1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	

1	2	3	4
5	6	7	8
9	10	11	12
13	15	14	

Artificial intelligence, database, circuit, algorithms

Topic: Number Theory

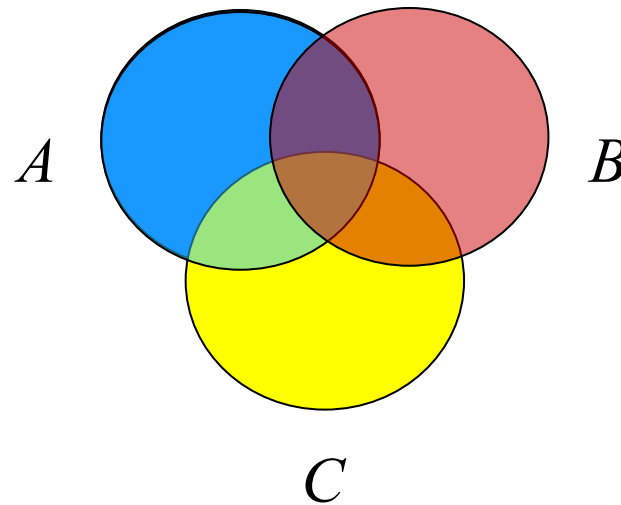
- (Extended) Euclidean algorithm and prime numbers
- **Modular arithmetic**, Chinese remainder theorem



Cryptography, coding theory, data structures

Topic: Counting

- Sets and Functions
- Combinations, Permutations, Binomial theorem
- Counting by mapping, pigeonhole principle
- **Recursions**



Probability, algorithms, data structures

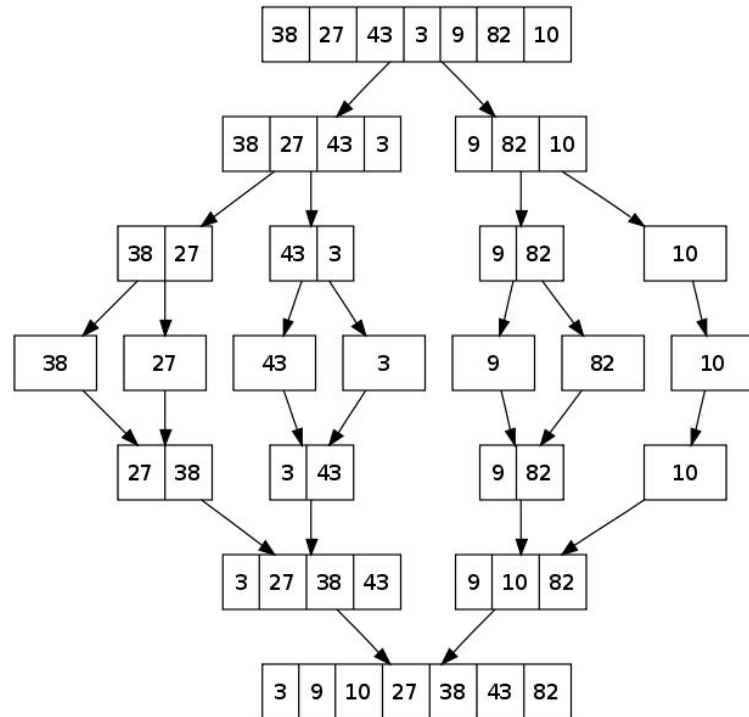
Topic: Counting

How many steps are needed to sort n numbers?

Algorithm 1 (Bubble Sort):

Every iteration moves the i -th smallest number to the i -th position

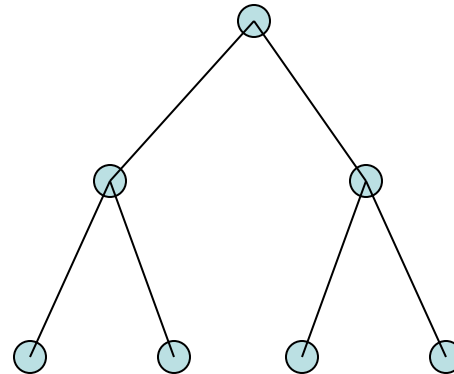
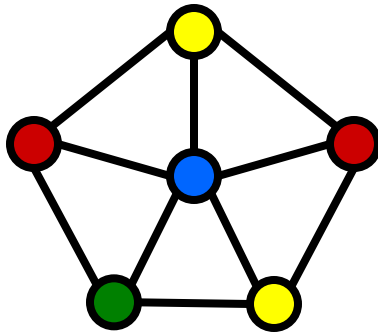
Algorithm 2 (Merge Sort):



Which algorithm runs faster?

Topic: Graph Theory

- Graphs, Relations
- Degree sequence, Eulerian graphs, isomorphism
- Trees
- Matching
- Coloring

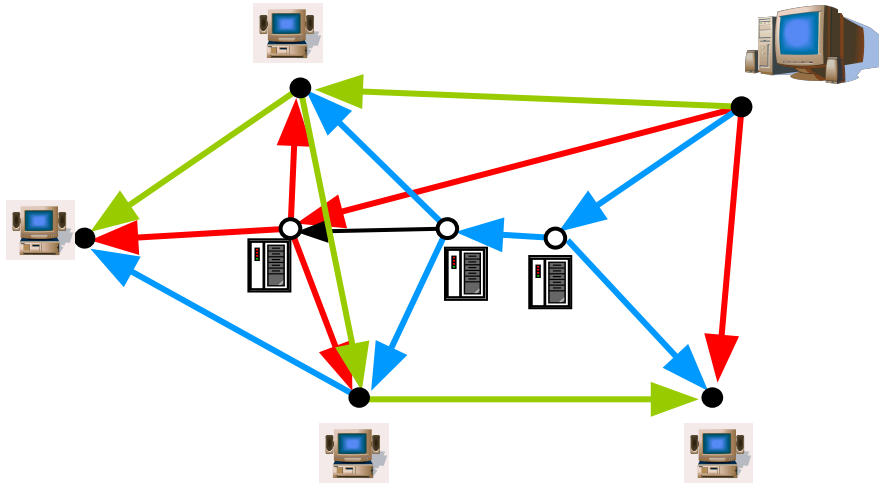
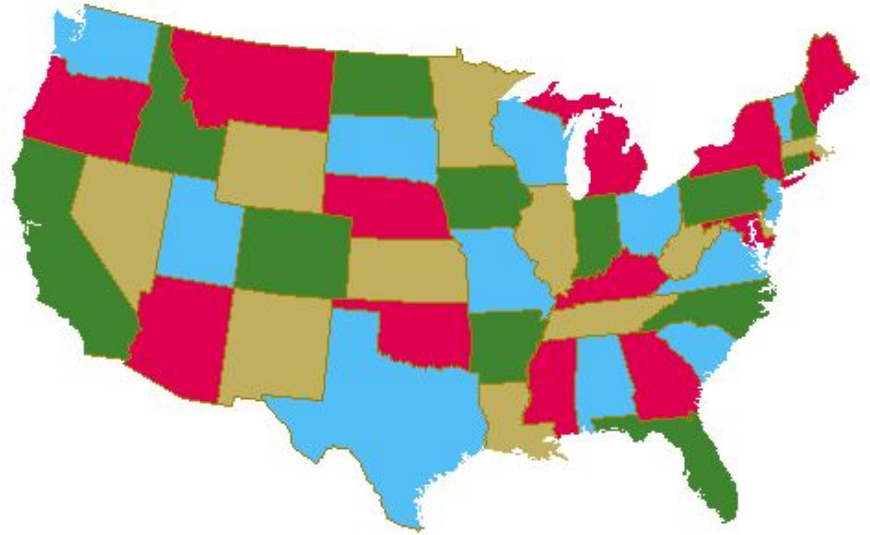


Computer networks, circuit design, data structures

Topic: Graph Theory

How to color a map?

How to schedule exams?



How to send data efficiently?

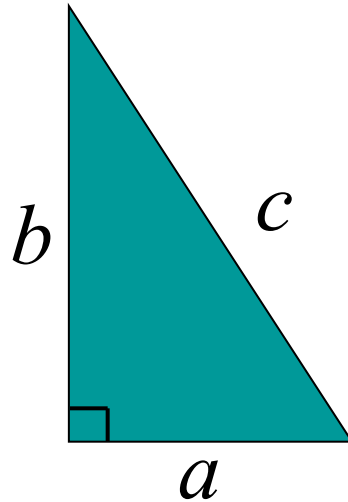
Objectives of This Course

- To learn basic mathematical concepts, e.g. sets, functions, graphs
- To be familiar with formal mathematical reasoning, e.g. logic, proofs
- To improve problem solving skills
- To see the connections between discrete mathematics and computer science

Practice, Practice and Practice.



Pythagorean theorem

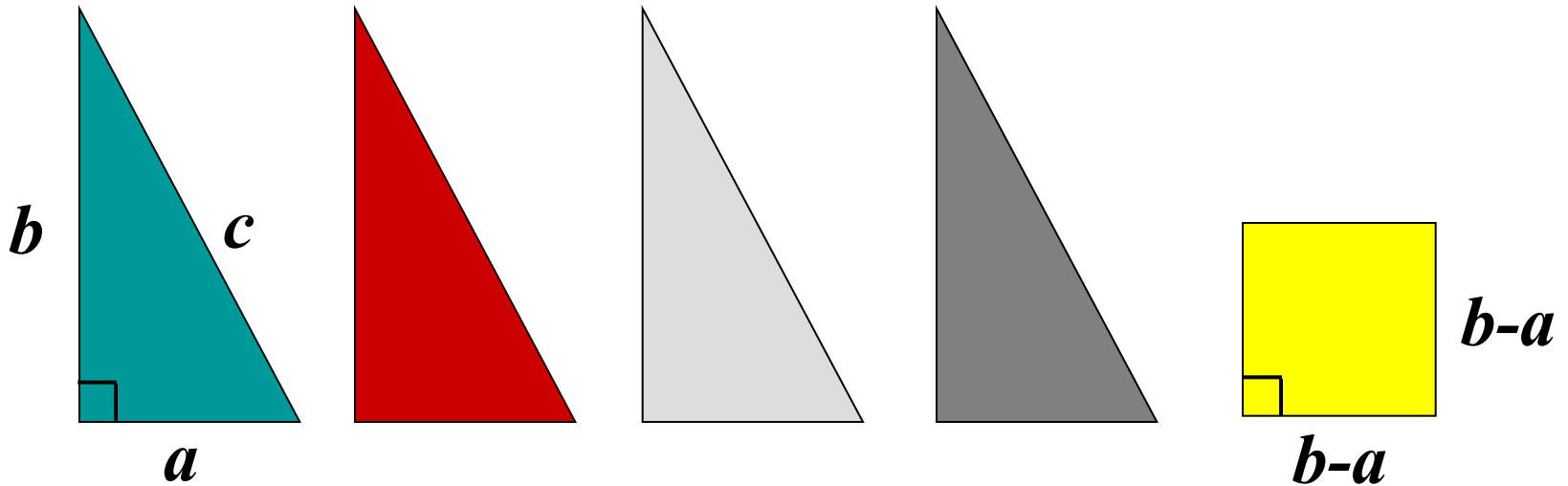


$$a^2 + b^2 = c^2$$

Familiar?

Obvious?

Good Proof



We will show that these five pieces can be rearranged into:

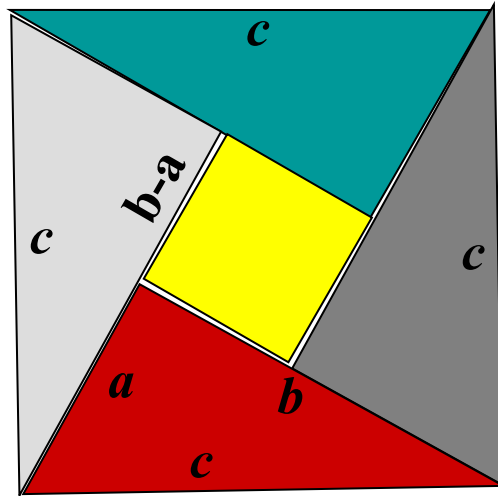
- (i) a $c \times c$ square, and then
- (ii) an $a \times a$ & a $b \times b$ square

And then we can conclude that $c^2 = a^2 + b^2$

Good Proof

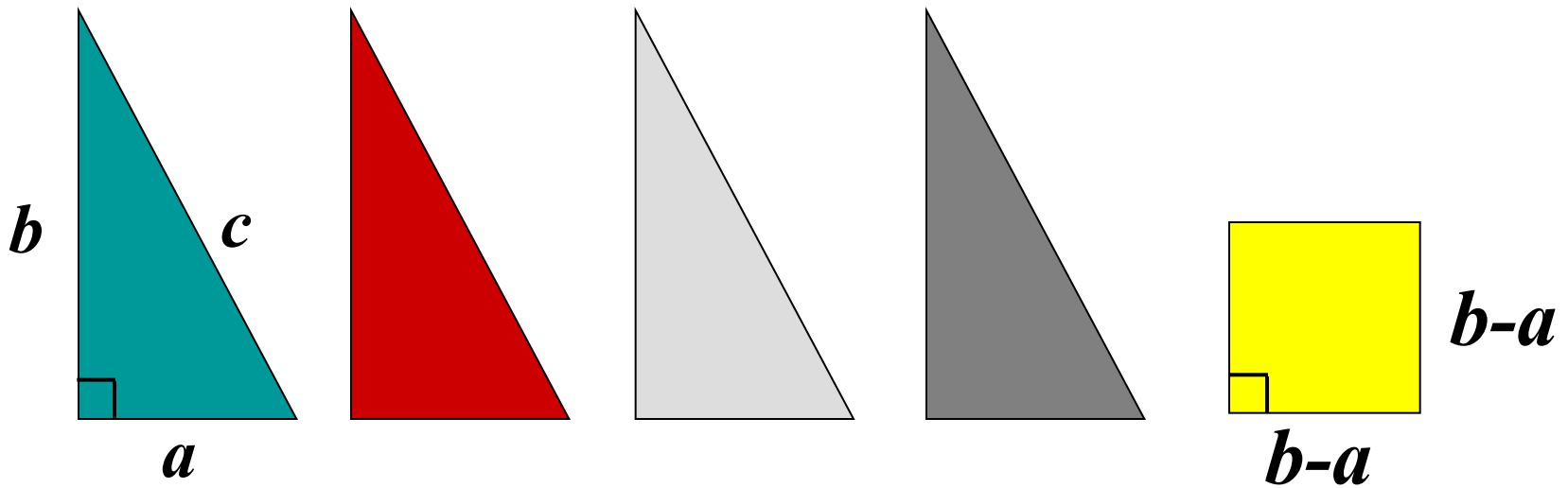
The five pieces can be rearranged into:

(i) a $c \times c$ square

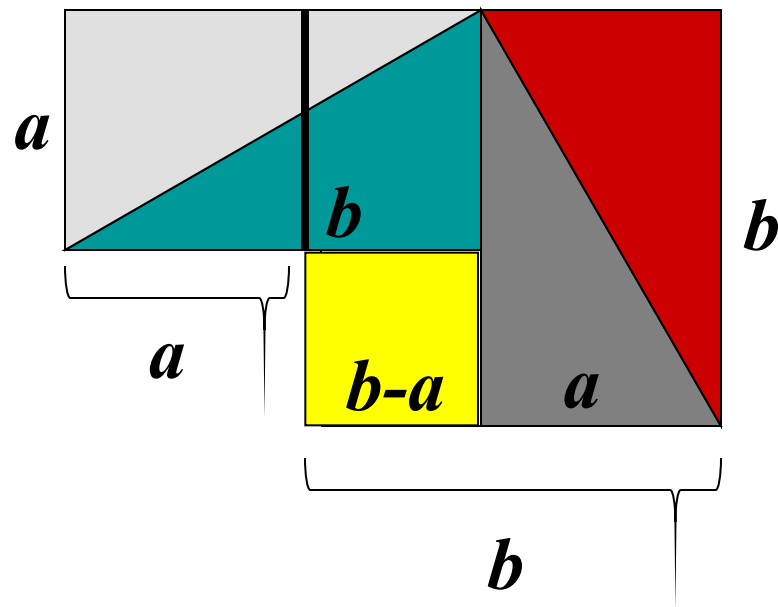


Good Proof

How to rearrange them into an $a \times a$ square and a $b \times b$ square?



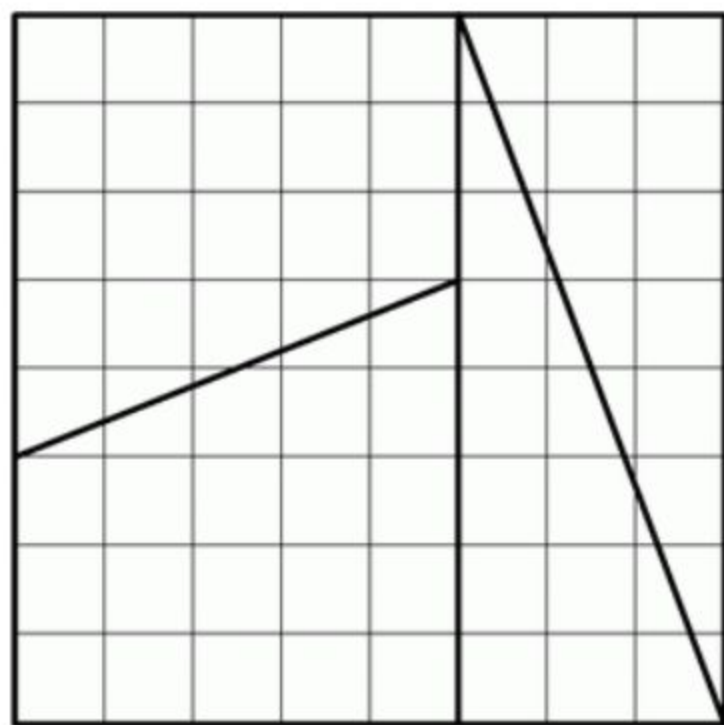
Good Proof



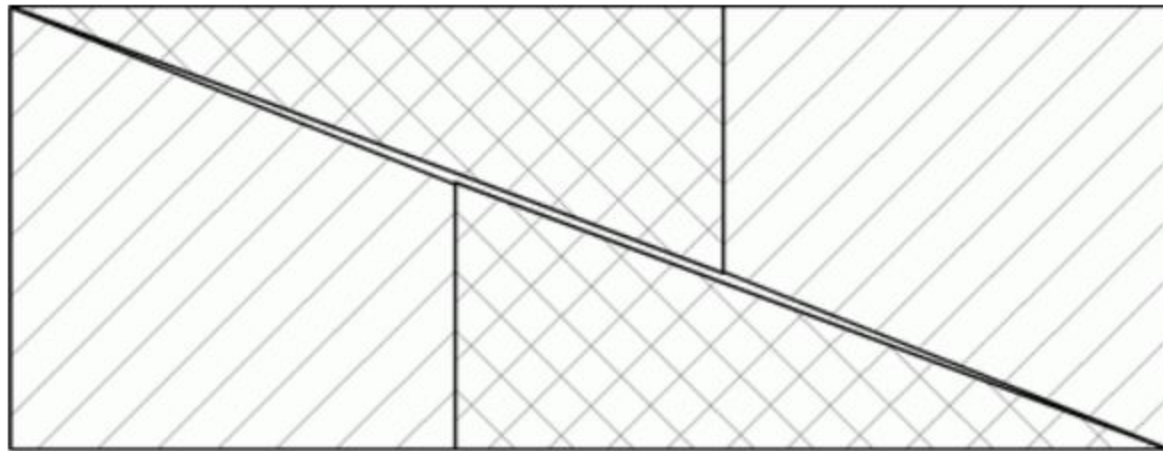
Bad Proof

A similar rearrangement technique shows that $65=64$...

What's wrong with the proof?



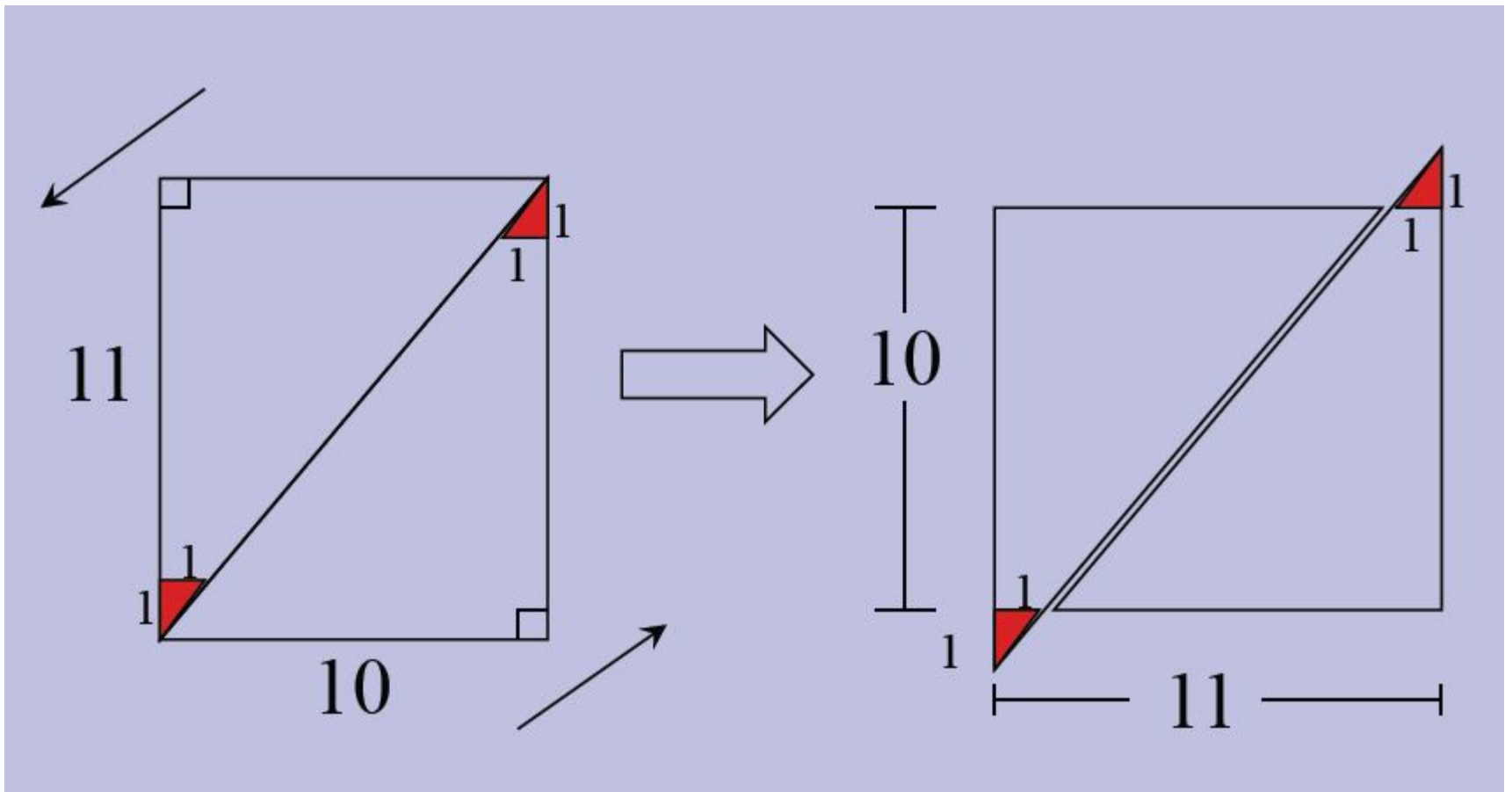
8×8



5×13

Bad Proof

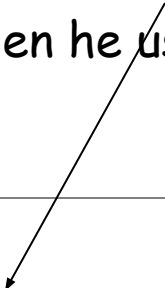
Another example



Mathematical Proof

To prove mathematical theorems, we need a more rigorous system.

The standard procedure for proving mathematical theorems is invented by Euclid in 300BC. First he started with five **axioms** (the truth of these statements are taken for granted). Then he uses **logic** to deduce the truth of other statements.

- 
1. It is possible to draw a straight line from any point to any other point.
 2. It is possible to produce a finite straight line continuously in a straight line.
 3. It is possible to describe a circle with any center and any radius.
 4. It is true that all right angles are equal to one another.
 5. ("Parallel postulate" ("Parallel postulate")) It is true that, if a straight line falling on two straight lines make the interior angles on the same side less than two right angles,
the two straight lines, if produced indefinitely, intersect the two straight lines, if produced indefinitely, intersect on that side on which are the angles less than the two right angles.



CONCEPTS



Statement (Proposition)

A *Statement* is a sentence that is either **True** or **False**

Examples:

$$2 + 2 = 4 \quad \text{True}$$

$$3 \times 3 = 8 \quad \text{False}$$

787009911 is a prime

Atomic statement:
if it cannot be divided
into smaller statements

Molecular statement:
otherwise

Non-examples:

$$x + y > 0$$

$$x^2 + y^2 = z^2$$

They are true for some values of x and y
but are false for some other values of x and y .

More Examples

Statements

- The moon is made of cheese.
- 42 is a perfect square.
- Every even number greater than 2 can be expressed as the sum of two primes.
- $3+7=12$

$$3 + x = 12, \text{ where } x = 9$$

$$3 + x = 12 \text{ for all values of } x$$

Non-Statements

- Would you like some cake?
- The sum of two squares.
- $1 + 3 + 5 + 7 + \dots + 2n + 1$.
- Go to your room!
- $3 + x = 12$



Logic Operators

$\neg ::= \text{NOT}$

$\sim p$ is true if p is false

$\wedge ::= \text{AND}$

$\vee ::= \text{OR}$

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

CONJUNCTION

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

DISJUNCTION

Compound Statement

Propositional
variable



It is hot and sunny

$$p \wedge q$$

It is not hot but sunny

$$\neg p \wedge q$$

It is neither hot nor sunny

$$\neg p \wedge \neg q$$

Without the definition of p and q , these are called statement form or propositional form

The truth table for a given statement form displays the truth values that correspond to all possible combinations of truth values of its component statement variables.

Compound Statement

Propositional
variable



p = "it will be cold"

q = "it will be rainy"

It will be cold or rainy

$p \vee$

It will either be cold or not rainy

q
 $p \vee$
 $\neg q$



Exclusive-Or

coffee "or" tea



exclusive-or

How to construct a compound statement for exclusive-or?

p	q	p ☐ q
T	T	F
T	F	T
F	T	T
F	F	F

Idea 1: Look at the true rows

Want the formula to be true exactly when the input belongs to a "true" row.

$$(p \wedge \neg q) \vee (\neg p \wedge q)$$

The input is the second row exactly if this sub-formula is satisfied

And the formula is true exactly when the input is the second row **or** the third row.

Exclusive-Or

coffee "or" tea



exclusive-or

How to construct a compound statement for exclusive-or?

p	q	p ☐ q
T	T	F
T	F	T
F	T	T
F	F	F

Idea 2: Look at the false rows

Want the formula to be true exactly when the input does **not** belong to a "false" row.

$$\neg(p \wedge q) \wedge \neg(\neg p \wedge \neg q)$$

The input is the first row exactly if this sub-formula is satisfied

And the formula is true exactly when the input is **not** in the 1st row **and** the 4th row.

Logical Equivalence

Idea 3: Guess and check

$$p \oplus q \equiv (p \vee q) \wedge \neg(p \wedge q)$$

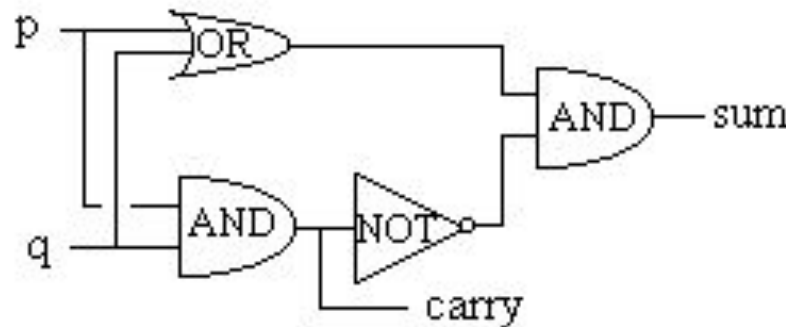
p	q	$p \oplus q$	$p \vee q$	$\neg(p \wedge q)$	
T	T	F	T	F	F
T	F	T	T	T	T
F	T	T	T	T	T
F	F	F	F	T	F

Logical equivalence: Two statements have the same truth table

As you see, there are many different ways to write the same logical formula. One can always use a truth table to check whether two statements are equivalent.

Writing Logical Formula for a Truth Table

Digital logic:



p	q	sum	carry
1	1	0	1
1	0	1	0
0	1	1	0
0	0	0	0

Given a digital circuit, we can construct the truth table.

Now, suppose we are given only the truth table (i.e. the specification), how can we construct a circuit (i.e. formula) that has the same function?

Writing Logical Formula for a Truth Table

Use idea 1 or idea 2.

Idea 1: Look at the true rows and take the "or".

	p	q	r	output
$p \wedge q \wedge r$	T	T	T	F
$p \wedge q \wedge \neg r$	T	T	F	T
$p \wedge \neg q \wedge r$	T	F	T	T
$p \wedge \neg q \wedge \neg r$	T	F	F	F
$\neg p \wedge q \wedge r$	F	T	T	T
$\neg p \wedge q \wedge \neg r$	F	T	F	T
$\neg p \wedge \neg q \wedge r$	F	F	T	T
$\neg p \wedge \neg q \wedge \neg r$	F	F	F	F

$$\begin{aligned} & (p \wedge q \wedge \neg r) \\ & \vee (p \wedge \neg q \wedge r) \\ & \\ & \vee (\neg p \wedge q \wedge r) \\ & \vee (\neg p \wedge q \wedge \neg r) \\ & \vee (\neg p \wedge \neg q \wedge r) \end{aligned}$$

The formula is true exactly when the input is one of the true rows.

Writing Logical Formula for a Truth Table

Idea 2: Look at the false rows, **negate** and take the **"and"**.

	p	q	r	output
$p \wedge q \wedge r$	T	T	T	F
$p \wedge q \wedge \neg r$	T	T	F	T
$p \wedge \neg q \wedge r$	T	F	T	T
$p \wedge \neg q \wedge \neg r$	T	F	F	F
$\neg p \wedge q \wedge r$	F	T	T	T
$\neg p \wedge q \wedge \neg r$	F	T	F	T
$\neg p \wedge \neg q \wedge r$	F	F	T	T
$\neg p \wedge \neg q \wedge \neg r$	F	F	F	F

$$\neg(p \wedge q \wedge r)$$

$$\wedge \neg(p \wedge \neg q \wedge \neg r)$$

$$\wedge \neg(\neg p \wedge \neg q \wedge \neg r)$$

The formula is true exactly when the input is **not** one of the false row.

DeMorgan's Laws

Logical equivalence: Two statements have the same truth table

Statement: Tom is in the football team and the basketball team.

Negation: Tom is not in the football team or not in the basketball team.

De Morgan's Law

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

Statement: The number 783477841 is divisible by 7 or 11.

Negation: The number 783477841 is not divisible by 7 and not divisible by 11.

De Morgan's Law

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

DeMorgan's Laws

Logical equivalence: Two statements have the same truth table

De Morgan's Law

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

p	q	$\neg(p \wedge q)$	$\neg p \vee \neg q$
T	T	F	F
T	F	T	T
F	T	T	T
F	F	T	T

De Morgan's Law

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$